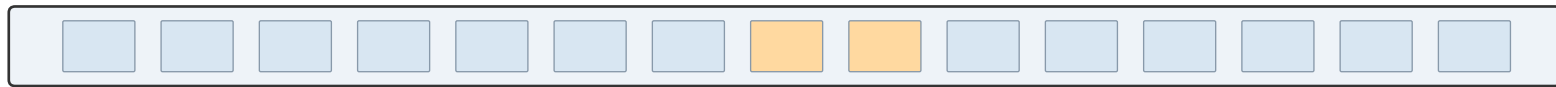


Frozen LLM, residual stream at layer ℓ $x_{i,j} \in \mathbb{R}^d$, $j = 1 \dots N$ (N up to 131,072)



k-token attack needle among benign context (dilution $k/N \approx 2.4 \times 10^{-4}$)

Softmax attention pooling

$\alpha_j = \text{softmax}_j(q^\top y_j / \sqrt{m})$, score tensor $(1, N)$
materialises $(1, N, m) \Rightarrow \Theta(N)$ memory

652.1 MiB overhead at $N = 131,072$ ($\alpha = 0.842$)

34×

MultiMax aggregation (ours)

$a_h = \max_j v_h^\top y_j$, H heads, streamed in chunks C
carries only (B, H) state $\Rightarrow \Theta(\min(C, N))$ memory

19.1 MiB overhead at $N = 131,072$ ($\alpha = 0.000$)

Cascade

MultiMax is the $\Theta(1)$ -memory first pass; the expensive monitor sees only what the gate cannot resolve.

Platt gate $\hat{p} = \sigma(az + b)$
escalate iff $|\hat{p} - 0.5| < \delta$

δ chosen from an escalation budget, not hard-coded

confident $\rightarrow$ probe verdict
(97.5% of traffic, $\Theta(1)$ memory)

ambiguous $\rightarrow$ LLM monitor
(2.5%, full inspection)